The cost of antibiotic resistance from a bacterial perspective

Johanna Björkman, Dan I. Andersson

Dept. of Bacteriology, Swedish Institute for Infectious Disease Control, Solna, Sweden

Abstract The emergence, spread and stability of antibiotic resistance in a bacterial population will be determined by several factors including (a) the volume of drug use, (b) the rate of formation of resistant mutants, (c) the biological cost of resistance and (d) the rate and extent of the genetic compensation of the costs. Generally, resistance is associated with a cost, suggesting that the frequency of resistant bacteria might decline when the use of antibiotics is decreased. However, evolution to reduce these costs, without a concomitant loss of resistance, can occur and result in a stabilization of the resistant bacteria in the population. The rate and trajectory of this compensatory evolution is dependent on the bacterial species, the specific resistance mutation and the environmental conditions under which evolution occurs.

INTRODUCTION

The introduction of antibiotics some 50 years ago was thought to herald the end of bacterial infectious diseases as a major cause of human mortality. However, this belief proved to be too optimistic. Of the many reasons for the resurgence of bacterial infection throughout the world, the rise of antibiotic resistance is particularly alarming and significant. Today antimicrobial resistance is an important health problem in both hospital- and community-acquired infections, and bacterial strains are now emerging that are resistant to all currently available antimicrobial agents.[1] Whether we can slow down, reverse or prevent this resistance development is unclear. A major obstacle is that our current knowledge is insufficient to describe and predict the development of antimicrobial drug resistance in a rational manner. This is mainly due to a lack of understanding of the different microbial, host and treatment parameters that determine the rates and trajectories of the emergence, spread and stability of resistance. This review summarizes our current understanding of some of the microbial factors that affect resistance development, in particular chromosomal resistance.

MICROBIAL FACTORS IMPORTANT FOR RESISTANCE DEVELOPMENT AND STABILITY

The rate and stability of resistance development is affected by several factors including (a) the volume of drug use, (b) the rate of formation of resistant mutants, (c) the biological cost of resistance and its genetic compensation.

Volume of drug use

The major selective pressure driving changes in the frequency of resistance is the volume of drug use. A number of pathogen-specific epidemiological models of drug resistance have been proposed for both community-acquired[2–4] and hospital-acquired infections.[5–8] These theoretical studies point out the volume of drug use as the major selective pressure driving changes in the frequency of resistance.

If the development of resistance is negatively associated with the bacteria's ability to survive, resistance will be eliminated once the selective pressure of the antibiotic is removed. There are, however, few clinical reports in support of the idea that a more prudent and reduced use of drugs will relax the selection pressure and be followed by a decline in the frequency of resistant bacteria in the community. In Iceland, the frequency of penicillin-resistant pneumococci increased rapidly in the late 1980s. Between 1992 and 1995 measures were taken to reduce antibiotic consumption in children and, probably as a result of this intervention, resistance peaked in 1993 at 19.8% and then declined and leveled off.[9] In Finland, after nationwide reductions in the use of macrolide antibiotics, there was a decline from 16.5% in 1992 to 8.6% in 1996 in the frequency of erythromycin resistance among group A *streptococci*.[10] Thus, for these cases a substantial reduction in antibiotic use resulted in a rather moderate decrease in the frequency of resistance that appeared to level off at a frequency much higher than before the increase. In contrast, both theoretical considerations and experimental studies suggest that a reduced antibiotic use in hospitals might lead to a more rapid and extensive decrease in the frequency of resistance than it would in the community.[8]

Rate of formation of resistant mutants

The rate of formation of resistant mutants will be determined by the bacterial mutation (or plasmid transfer;[11]) rates and the population size. In a population where the rate of evolution is limited by mutation supply, a high mutation rate might act to increase the rate of resistance development.[12] Thus, mutator bacteria with increased mutation rates and recombination frequencies could contribute to the emergence of antibiotic resistance. This can either be due to the increased mutation rate in indigenous genes or to the increase in frequency of interspecies recombination of foreign antibiotic-resistance genes acquired by horizontal gene transfer. A few per cent of natural bacterial isolates of *Escherichia coli* and *Salmonella enterica* are mutator clones that have high mutation rates[13] whereas in *P. aeruginosa* about one third of the isolates from cystic fibrosis patients show a mutator phenotype.[14] Antibiotic selection can greatly increase the frequency of mutator bacteria in a population and the number of mutators can be further enhanced by additional selections.[15]

The apparent mutation rate can also vary due to changes in the physiological or selective conditions. For example, bacteria can use gene amplification to adapt to selective conditions.[16] The chromosomal β-lactamase gene *ampC* in *E. coli* has been shown to amplify in response to penicillin selection.[17] The gene copy number is proportional to the amount of β-lactamase formed and the ampicillin resistance is

doi: 10.1054/drup.2000.0147, available online at http://www.idealibrary.com on IDEAL®

proportional to the level of enzyme produced. Likewise, an increased number of copies of the target gene stabilized in the presence of the selective agent increases the number of mutable targets and will thereby increase the apparent mutation rate, as described by Andersson et al.[16] The mechanism described could explain how penicillin-resistant strains rapidly change substrate specificity and become resistant to the newer cephalosporins.

Costs of resistance and genetic compensation

A third factor of great importance is the fitness costs of resistance and the ability of the bacteria to genetically compensate for such fitness costs (Fig. 1). The fitness of an antibiotic-resistant pathogen is determined by the relative rates at which resistant and sensitive bacteria (i) grow and die in hosts and environment, (ii) are transmitted between hosts, and (iii) are cleared from infected hosts.[18] To accurately describe and predict the fitness of resistant bacteria we need to experimentally quantify all these parameters.

Genetic compensation of the fitness costs of resistance can occur through (a) reversion or loss of the resistance gene/element, (b) genetic changes in the element responsible for resistance and/or host adaptation to carriage of this element, and (c) intra- or extra-genic suppression of the chromosomal resistance gene. Fixation of the compensatory mutation in the population is affected by the mutation rate to compensation, the fitness of the compensated mutant and the effective population size.

Even though transmission and clearance costs can be measured, most experimental estimates of the cost have focussed on measurements of the growth performance of the resistant mutants. In these assays the relative fitness of antibiotic-sensitive and -resistant bacteria is measured during competition experiments in either laboratory media (chemostats, batch cultures), sterilized soil, cell cultures or in experimental animals. During competition, sensitive and resistant bacteria are followed over time and changes in the relative frequencies are followed by selective plating. The competition index (C.I.) is calculated as the ratio of (antibiotic-resistant bacteria/antibiotic-sensitive bacteria) at time (t_1)/(antibiotic-resistant bacteria/antibiotic-sensitive bacteria) at time (t_0). Differences in C.I. can be caused by variation in any of the components of growth fitness: lag phase, rates of exponential growth, resource utilization efficiencies, and rates of mortality in the presence and absence of host defences. These pairwise competition experiments can detect differences in fitness at least as small as 0.5%.

At present we do not know how well cost measurements in the laboratory reflect costs in the real world. It is likely that if the resistance mutation confer a cost under some condition in the laboratory it will also confer a cost in a natural population. However, if no fitness cost is observed under a variety of laboratory conditions it still remains possible that there are natural conditions where the costs are substantial.

The rate of the amelioration of cost can be measured experimentally by following the growth of the resistant strain (in the absence or presence of antibiotic) in chemostats, by serial transfer in laboratory cultures, cultured cell-lines or in laboratory animals. Thus, slow-growing resistant bacteria are selected for faster growth/enhanced fitness,

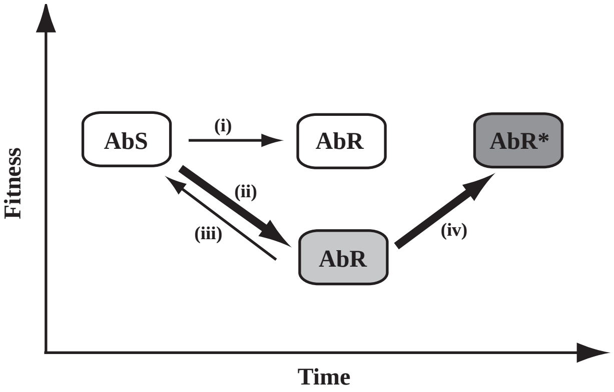

Fig. 1 **Stability of antibiotic resistance in bacteria. Antibiotic resistant bacteria (AbR) may be as fit (i) or less fit (ii) than their sensitive variants. Fitness may be restored (iii) by true reversion of the resistance conferring mutation or (iv) by acquisition of compensatory mutations that restore fitness to different extent without causing loss of resistance (AbR*).**

sampled after serial transfer and plated to detect the number of fast-growing compensated mutants in the population. The degree of compensation can be measured by pairwise competition with the ancestral-resistant mutant or a neutrally tagged reference strain.

WHAT ARE THE COSTS OF RESISTANCE AND HOW ARE THEY AMELIORATED?

Retrospective studies

Levin et al.[2] and Austin et al.[4] illustrate two different mathematical models of the population genetics and epidemiology of antibiotic resistance. These theoretical studies provide the information needed to identify and quantitatively evaluate the role of different factors contributing to the ascent and maintenance of drug resistance, and to interpret and evaluate the results of intervention strategies. In addition, the fitness cost of resistance can be indirectly estimated from these mathematical models by determining which cost best predicts the observed decrease in the frequency of resistant bacteria in natural populations in response to reduced drug consumption.[4] There are, however, few epidemiological studies that have recorded temporal changes in both the frequency of resistance to a specific drug and the volume of drug consumption in the community, making these types of inferences about costs difficult.[9,10,19]

Prospective studies

At present, even though these studies are urgently needed, no prospective studies on the cost of resistance have been performed. Individual patients infected with either sensitive or resistant bacteria should be followed over time as the rate of loss of the strains is recorded. From such experiments we can measure the effect of resistance on fitness in individual patients and determine a relationship between resistance and drug use within individual patients. In addition, controlled competition experiments between isogenic sensitive

Drug Resistance Updates (2000) 3, 237–245

and resistant bacteria performed on human subjects would be of great interest.

Experimental studies

The cost of chromosomal resistance

Resistance caused by chromosomal mutations is primarily achieved by the modification of target molecules. These chromosomal genes are responsible for essential cellular activities and it is reasonable to assume that a resistance-conferring mutation may affect the growth rate and reduce fitness. For example, mutations in the *rpoB* gene encoding the β subunit of RNA polymerase responsible for rifampicin resistance in *E. coli* can diminish transcriptional efficiency and reduce the growth rate.[20] Mutations in the *rpsL* gene encoding ribosomal protein S12 can cause a decreased elongation rate and a depressed growth rate.[21] Single amino acid substitutions in the *fusA* gene encoding elongation factor G (EF-G) cause fusidic acid resistance in *S. typhimurium*. Most mutations interfere with conformational changes of EF-G on the ribosome, which in turn affect translation elongation rate.[22]

Most studies of the cost of resistance performed on resistance due to target alterations show some fitness cost to resistance (Table 1). We have experimentally measured the costs of resistance in *S. typhimurium* and *S. aureus* in experimental animal models and in laboratory medium. The virulence of 17 *S. typhimurium* strains with high level resistance to streptomycin (SmR), rifampicin (RifR), nalidixic acid (NalR) or fusidic acid (FusR) and their isogenic sensitive wild-type strain was measured in competition experiments in mice.[32] As summarized in Figure 2, the wild type outcompeted almost all resistant *S. typhimurium* mutants. Only one

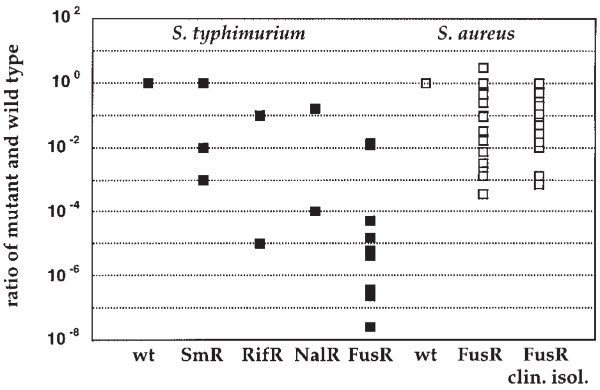

Fig. 2 **Virulence of wild type and antibiotic-resistant *S. typhimurium* (filled squares) assayed in a mouse model for typhoid fever and *S. aureus* (open squares) measured in a rat model for abscess formation.**

SmR strain was as virulent or slightly more virulent than the wild type. The variation in competition index (ratio of mutant and wild type) ranged from 5 to 10^{-7}. Among the resistant strains, the FusR mutants were shown to be most avirulent compared to the wild type. The cost of resistance was also measured in vitro; 14 out of the 17 resistant mutants had a slower growth rate than the sensitive wild-type strain in LB medium. The virulence of *S. aureus* mutants resistant to fusidic acid was measured in a rat model for wound infections. Figure 2 shows that the majority of the resistant mutants were less virulent than the sensitive wild type. All FusR mutants examined also had slower growth rates in LB medium.

Table 1 **The biological cost of chromosomal resistances**

Bacteria	Resistance	Mutation	Cost	Assay system	Reference
S. typhimurium	Streptomycin	*rpsL*	yes/no	mice, in vitro	32,36
	Rifampicin	*rpoB*	yes	mice, in vitro	32
	Nalidixic acid	*gyrA*	yes	mice, in vitro	32
	Fusidic acid	*fusA*	yes/no	mice, in vitro	22,37
E. coli	Streptomycin	*rpsL*	yes/no	in vitro	29,31
	Rifampicin	*rpoB*	yes/no	in vitro	20,24
M. tuberculosis/bovis	Isoniazid	*katG*	yes	mice, guinea pigs	26,28,34
	Rifampicin	*rpoB*	yes/no	in vitro	35
S. aureus	Fusidic acid	*fusA, non-fusA*	yes/no	rats, in vitro	Nagaev et al., unpublished
	Rifampicin	*rpoB*	yes/no	in vitro	Björkman et al., unpublished
H. pylori	Clarithromycin	23SrRNA	yes	mice	Björkholm et al., unpublished
P. aeruginosa	Ciprofloxacin	?	yes	in vitro	Björkman, unpublished
	Gentamicin	?	yes	in vitro	Björkman, unpublished
	Rifampicin	?	yes	in vitro	Björkman, unpublished
P. fluorescens/putida	Rifampicin	?	yes	soil	23
B. subtilis	Rifampicin	*rpoB*	yes/no	in vitro	25
L. monocytogenes	class IIa bacteriocin	?	yes	in vitro	33
N. meningitidis	Sulfonamide	*sul2*	yes	in vitro	30

Table 2 **The biological cost of resistance encoded by extrachromosomal elements**

Bacteria	Plasmid (resistance)	Cost	Assay system	Reference
E. coli K12	R1 (KmR, ApR, CmR)	yes	C-limited chemostat	11
E. coli K12	R (vary)	yes/no	Colombia broth culture	39
E. coli K12	TP120 (TcR, ApR, SmR, SuR)	yes	C-, or P-limited chemostat	38
E. coli HBT	pBR322 derivative (ApR, TcR)	yes/no	LB broth culture	50
E. coli K12	pBR322 (ApR, TcR)	yes	Minimal medium batch culture	47
E. coli JA104	pBR322 (ApR, TcR)	yes	C-limited chemostat	45
E. coli	pBR322 (ApR, TcR)	yes	C-limited chemostat	46
E. coli RR1	pBR322 (ApR, TcR)	yes	C-limited chemostat/batch	41
E. coli	pBR322::Tn3 (ApR)	yes	C-limited chemostat	48
E. coli B	pACYC184 (TcR, CmR)	yes	C-limited minimal medium culture	49
E. coli B	pACYC184 (TcR, CmR)	yes	C-limited minimal medium culture	42
E. coli	RSF2124 (ApR)	yes	C-limited chemostat	40
E. coli	R388:Tn1721 (HgR, TcR)	yes	sterile well water culture	43
P. cepacia	R388:Tn1721 (HgR, TcR)	yes	sterile well water culture	43
K. pneumoniae	R388:Tn1721 (HgR, TcR)	yes	sterile well water culture	43
E. agglomerans	R388:Tn1721 (HgR, TcR)	yes	sterile well water culture	43
E. coli	pRO101 (TcR, TpR)	yes	sterile well water culture	43
P. putida	pRO101 (TcR, TpR)	yes	sterile well water culture	43
B. subtilis	pBR322-pUB110 (ApR, NmR)	yes	in vitro	44

Ap (ampicillin), Tc (tetracycline), Sm (streptomycin), Su (sulfonamide), Km (kanamycin), Cm (chloramphenicol), Hg (hygromycin), Nm (neomycin) Tp (trimetoprim).

However, mutants with no cost have also been observed. An example of a no-cost resistance mutation is the AAA_{42} (Lys) -> AGA_{42} (Arg) substitution of the *rpsL* gene, conferring high level streptomycin resistance in *E. coli*[29,31] and *S. typhimurium*.[32] High level rifampicin resistance caused by the following substitutions of the *rpoB* gene TCG_{531} (Ser) -> TTG_{531} (Leu) in *M. tuberculosis*,[35] CAT_{481} (His) -> TAT_{481} (Tyr) in *S. aureus* (Nagaev et al., unpublished) and GAC_{516} (Asp) -> GGC_{516} (Gly) in *E. coli*[20] also appear not to confer any cost.

The cost of resistance can not be predicted at present from the type of target molecule or from the nature of the specific mutation. The streptomycin resistant *S. typhimurium* mutants AAA_{42} (Lys) -> AAC_{42} (Asn) and AAA_{42} (Lys) -> ACA_{42} (Thr) cause severe fitness costs whereas AAA_{42} (Lys) -> AGA_{42} (Arg) is selectively neutral or slightly advantageous in mice. Thus, to obtain a complete picture of the cost of resistance towards a given antibiotic it is necessary to examine several types of mutations in a given target, as well as different targets if resistance can be conferred by different mechanisms.

The cost of plasmid-encoded resistance

Resistance caused by plasmids, phages and transposons is generally due to the introduction of enzymes that inactivate the antibiotic or pumps that remove it from the cell. A number of studies have reported fitness burdens associated with the carriage of resistance-encoding (R) plasmids (Table 2). Some studies have shown that the resistance determinant per se conferred the cost to the host, but extrachromosomal-element-encoded resistance may also give rise to costs associated with replication and maintenence of the elements themselves. The majority of the studies listed in Table 2 have examined naive associations between bacteria and plasmids, meaning there is no evolutionary history of association between plasmid and host genotype used. In addition, most studies of plasmids have been done in vitro and with cloning vectors or plasmids that have been maintained in the laboratory for long periods of time. In contrast, the costs of resistance for different plasmid-host combinations from natural populations have not been examined.

Amelioration of fitness costs by compensatory evolution

Chromosomal resistance

Evolution to reduce the cost of resistance by compensation occurs both in the presence and absence of drug selection in laboratory media, experimental animals and in humans. In the presence of drugs obviously resistance has to be maintained and only those compensatory mutations that do not reduce resistance are selected. However, even in the absence of drugs, compensatory second-site mutations that partly or fully restore fitness and retain the resistance are more common than reversion to drug sensitivity (Table 3).

We have studied the genetic compensation of the cost of resistance in *S. aureus* and *S. typhimurium* strains by allowing resistant strains with low fitness to evolve by serial passage in the absence of antibiotics in experimental animals and laboratory medium. In *S. typhimurium*, fast-growing variants appeared from the avirulent resistant SmR, RifR, and NalR mutants after two to six growth cycles in mice. Of 28 independent compensated mutants isolated from two SmR mutants in mice and laboratory medium, 27 compensated mutants maintained high-level Sm resistance. The compen-

Table 3 **Genetic compensation of fitness cost caused by chromosomal mutations**

Bacteria	Resistance mutation (resistance)	Compensatory mutation (resistance in compensated mutant)	Selection for compensation in	Reference
S. typhimurium	*rpsL* (streptomycin)	intragenic, *rpsL* (maintained)	mice	32
	rpsL (streptomycin)	extragenic, *rpsD/E* (maintained)	laboratory medium	32
				37
	gyrA (nalidixic acid)	intragenic, *gyrA* (maintained)	mice	32
	rpoB (rifampicin)	intragenic, *rpoB* (maintained)	mice	32
	fusA (fusidic acid)	true reversion, *fusA* (lost)	mice, laboratory medium	22,37
	fusA (fusidic acid)	intragenic, *fusA* (often maintained)	laboratory medium	22,37
S. aureus	*fusA* (fusidic acid)	extragenic, ? (maintained)	rats	Nagaev et al., unpublished
	fusA (fusidic acid)	intragenic, *fusA* (often maintained)	laboratory medium	Nagaev et al., unpublished
	rpoB (rifampicin)	intragenic, *rpoB* (often maintained)	laboratory medium	Björkman et al., unpublished
E. coli	*rpsL* (streptomycin)	extragenic, *rpsD/E* (maintained)	laboratory medium	29,31
	rpoB (rifampicin)	intragenic, *rpoB* (maintained)	laboratory medium	20
M. tuberculosis	*katG* (isoniazid)	extragenic, *ahpC* (maintained)	humans	51
N. meningitidis	*sul2* (sulfonamide)	intragenic, *sul2* (maintained)	humans	30

sated mutants partly or fully regained their ability to compete with the wild type in mice as well as in laboratory medium. In *S. aureus*, fast-growing variants appeared from avirulent resistant FusR mutants in rats and laboratory medium; 45 out of 47 fast-growers maintained their high resistance level. We conclude from these studies that true reversion is much less common than compensatory evolution and that most compensated mutants maintain high-level resistance.

The fact that compensatory mutations are more common than reversion can be explained by the higher rates of compensatory mutations, relative to reversions, and the population bottlenecks associated with serial passage.[54] Thus, even if compensated mutants might have a lower fitness than revertants, they can still become fixed if they form more frequently and population bottlenecks are present. Furthermore, a reversion to sensitivity is unlikely to occur once the resistant and compensated mutant has become fixed in the population. Thus, results from studies of streptomycin resistance in *E. coli*[29,31] and *S. typhimurium*[32,36,37] show that the extragenic compensatory mutations alone confer a selective disadvantage. Consequently, when compensation has occurred it becomes difficult to get rid of the resistance mutation even in the absence of antibiotic selection pressure.

Fortunately there are a few exceptions where reversion to sensitivity is more common than second-site compensation. For example, sensitive revertants are frequently found among fusidic acid-resistant *S. typhimurium* mutants that evolve in experimental animals. Thus, it would be expected that fusidic acid resistance cannot be stabilized by compensation but instead reversion to sensitivity is more likely to occur in the absence of antibiotic pressure. This example illustrates the importance of cost-compensation studies to identify resistances where amelioration of cost occurs by true reversion.

An interesting observation is that relatively often the compensated mutants have a higher fitness than the original sensitive parent strain. This type of 'over compensation' has been found during evolution in vitro,[20,49] in experimental animals[32] as well as in human patients (Björkholm et al., unpublished). From an evolutionary point of view this finding suggests that the use of antibiotics and the development of antibiotic resistance might allow a bacterial population to pass an adaptive valley and go from one adaptive peak to another at a rate that is unlikely in the absence of antibiotics. Since these phenomena can cause an increased fitness of the evolved clones they might have a clinical relevance by altering/increasing the virulence of the evolved resistant clones and alter the pathogenesis of the disease.

The physiological mechanism by which compensatory mutations restore fitness has been determined in a few cases. The cost of streptomycin resistance *rpsL* mutations in *E. coli* and *S. typhimurium* can be compensated for by extragenic *rpsD* or *rpsE* mutations that restore the efficiency and rate of translation to wild type or nearly wild-type levels.[29,31,36] Likewise, intragenic compensatory mutations of the *rpoB* gene restore the rate of transcription to wild-type levels in rifampicin resistant *E. coli*[20] and extragenic *ahpC* suppressor mutations are likely to restore catalase activity and fitness in *katG* isoniazid resistant *M. tuberculosis*.[51]

The nature of compensatory changes that evolve is dependent on the environment.[37] Compensatory evolution in resistant *S. typhimurium* was different within and outside the host. Three avirulent mutants with high levels of resistance to streptomycin or fusidic acid, respectively, were evolved. The types of compensated mutants obtained in mice were significantly different from those obtained in laboratory medium. For example, all compensated SmR mutants selected from mice had acquired intragenic suppressor mutations in the *rpsL* gene whereas all mutants selected from laboratory medium had extragenic suppressor mutations in the *rpsD* or *rpsE* genes. The compensated SmR mutants retained their high resistance level and increased their fitness in both mice and laboratory medium. Thus, the difference in mutation spectrum could not be explained by a

Table 4 **Genetic compensation of fitness costs caused by extrachromosomal elements**

Bacterial host	Plasmid (resistance)	Compensatory changes on	Selection for compensation in	Reference
E. coli K12	pTP120 (ApR, TcR, SmR, SuR)	plasmid (loss of TcR or ApR)	C- or P-limited chemostat	38
E. coli B	pACYC184 (TcR, CmR)	chromosome	C-limited batch culture	42
E. coli B	pACYC184 (TcR, CmR)	chromosome	C-limited batch culture	49
E. coli B	pBR322 (TcR, ApR)	chromosome	C-limited batch culture	
E. coli K12	pBR322 (TcR, ApR)	chromosome	minimal-salt batch culture	47
E. coli RH202	RSF2124 (ApR)	chromosome	C-limited chemostat	40
E. coli RR1	pBR322 (TcR, ApR)	chromosome (deletion of TcR)	C-limited batch culture/chemostat	41
E. coli	pBR322::Tn3 (ApR)	chromosome (Tn3 transposition)	C-limited chemostat	48
E. coli	pBR322 (TcR, ApR)	plasmid (loss of TcR)	C-limited chemostat	46
E. coli JA104	pBR322Δ5 (TcR, ApR)	chromosome and plasmid	C-limited chemostat	45

Ap (ampicillin), Tc (tetracycline), Sm (streptomycin), Su (sulfonamide) Cm (chloramphenicol).

poor fitness compensation since there was no difference in competitive ability of the *rpsD* extragenic compensated mutants and the intragenic *rpsL* compensated mutants.

For the FusR mutants, almost all compensated mutants obtained from laboratory medium had intragenic suppressor mutations in the *fusA* gene whereas most bacteria that had evolved in mice were compensated by true reversion of the resistance-conferring mutation. Thus, laboratory medium selection resulted in compensated resistant mutants whereas selection in a host favored evolution towards reversion and antibiotic sensitivity. Furthermore, the laboratory-selected compensated mutants were as fit as the wild type in laboratory medium but showed a severe fitness decrease in mice whereas the mouse-selected revertants had restored their fitness in mice and laboratory medium. Thus, a difference in selection could account for the different mutation spectra. The environmental dependence of costs and evolution to reduce costs strongly supports the view that experimental estimates of the costs and stability of resistance should be done under a variety of culture conditions, in particular in experimental animals, but also, if possible, in humans to obtain the most relevant data.

Measurements of *S. typhimurium* mutation frequencies during growth in a host indicate that the mutation rate might be higher than in laboratory medium. Thus, the rate of pathogen adaptation might be faster in a host, which obviously could be important for the rate of resistance development and compensatory evolution.[37] One potential explanation for the increase in mutation frequency might be that growth in macrophages expose the bacteria to mutagenic compounds, e.g. oxygen and nitroso radicals. Nitric oxide (NO), superoxide and their reaction product peroxynitrite are generated excessively during host response against microbial infections. Peroxynitrite has been shown to increase the mutation rate of Sendai virus. Furthermore, the Sendai virus mutation rate was four times higher in wild-type mice compared to mice deficient in NO production,[52] suggesting that NO is indeed acting as a mutagen on invading pathogens. Thus, if oxidative stress induced by NO during the natural course of infection can accelerate bacterial mutation rates, it might be important for the pathogen in its adaptations to the host immune system, within-host competition with other bacteria, as well as for antibiotic resistance development.

Clinical data

Chromosomal resistance

An important question is how common and relevant compensation is in the real world. At present there is not enough clinical data available to answer this question but since compensation is frequently observed in laboratory media and in experimental animals, there is no reason to assume it will not occur in humans. From a theoretical point of view it is clear that the resistant clones which will appear and dominate in the population is determined by the respective mutation rates, the population size and the fitness of the mutants. Thus, if the population size and mutation rates are high enough to allow all resistant mutants to be present in the population, it is likely that the most fit mutant will dominate clinically. However, this can be altered by genetic bottle necks and/or variation in the specific mutation rates, resulting in fixation of less fit mutants.[37,54] Thus, if a less fit resistant mutant ascend to a high frequency in the population, evolution to reduce these costs, either by compensatory mutations or true reversion, is likely to occur.

One approach to examine the relevance of the experimental studies is to determine if the resistance mutations in bacteria isolated from humans are the same as those found in experimental studies. From the few studies performed there is evidence both for selection of 'no cost' mutants and genetic compensation of low fitness mutants. Results from one study suggest that the chromosomal mutations causing resistance in clinical isolates are likely to be the same as those that have a low cost experimentally.[53] Thus, streptomycin resistant clinical isolates of *M. tuberculosis* harbor the *rpsL* mutation that was shown not to confer any cost in *E. coli*[29,31] and *S. typhimurium*.[32,36,37] In contrast, in a study by Sherman et al.[51] there was good evidence for compensation. Here, isoniazid resistant *M. tuberculosis* isolates obtained from humans were examined. Resistance to isoniazid is often caused by knockout mutations in the *katG* gene which cause a loss of catalase activity and render the bacteria avirulent in mice.[26,28,34] Most of the catalase defective clinical isolates examined by Sherman et al. contained promoter-up mutations in the *ahpC* gene which cause increased production of alkyl hydroperoxidase reductase. This increase most likely compensated for the lack of catalase activity and reduction in virulence in the isoniazid resistant mutants.

Plasmid resistance

The fitness cost associated with plasmid carriage can also be reduced or eliminated by allowing the host-plasmid to coevolve. The results suggest that compensatory adaptation to extrachromosomal elements occur through mutations in the bacterial chromosome and/or in the plasmid itself (Table 4). Five evolutionary changes are possible in response to the carriage of a costly plasmid according to Modi and Adams.[45] First, adaptive genetic changes can occur in the bacterial chromosome that are independent of the presence of the plasmid. Second, there can be adaptive genetic changes in the bacterial chromosome that ameliorate the deleterious effect of the plasmid. Third, adaptive genetic changes in the plasmid can render the plasmid competitively superior in relation to other plasmids within the cell. Fourth, adaptive genetic changes in the plasmid can result in a general amelioration of the deleterious effect of the plasmid. Fifth, there may be adaptive genetic changes in both the bacterial chromosome and the plasmid which are specific to each other.

IMPLICATIONS

A general, depressing implication from the types of studies presented here is that resistance is likely to be with us once it has established itself in a bacterial population. One reason for this is the selection and establishment of 'no cost' resistant mutants. A second reason is that compensatory evolution can ameliorate the costs of resistance in low fitness mutants, without a concomitant loss of resistance. Both of these phenomena will result in stabilization of the resistant clones in the population. However, the more positive conclusion is that we can use this type of data in designing new antibiotics. Initially, the limiting factor in resistance development is the rate of mutation to resistance or the rate of plasmid transfer. But a low rate of mutation or plasmid transfer will only delay the initial appearance of resistance. The geographical spread and survival of the resistant pathogen is more coupled to the fitness costs of resistance and the environmental drug selection pressure. Knowledge of the cost of resistance and the risk of appearance of compensatory mutations can therefore be used for a rational design of antibiotics, in particular for resistance caused by chromosomally encoded target alterations. Thus, the ideal antibiotic would be a drug for which resistance mutations are rare, the fitness costs of resistance high and the rate of fitness-restoring compensatory mutations low. Hopefully, the relevance of cost and compensation for the rate of resistance development will be realized during future antibiotic design.

Acknowledgements

This work is supported by the Swedish Institute for Infectious Disease Control, Swedish Medical and Natural Science Research Councils, Swedish Strategic Research Foundation and Leo Pharmaceuticals Inc.

Received 20 June 2000;
Revised 3 July 2000;
Accepted 3 July 2000

Correspondence to: Dan I. Andersson, Dept. of Bacteriology, Swedish Institute for Infectious Disease Control, Nobels väg 18, 171 82 Solna, Sweden. Tel: +46 8 457 2432; Fax: +46 8 301 797; E-mail: Dan.Andersson@smi.ki.se

References

1. McCormick JB. Epidemiology of emerging/re-emerging antimicrobial-resistant bacterial pathogens. Curr Opin Microbiol 1998; 1: 125–9.
2. Levin BR, Lipsitch M, Perrot V et al. The population genetics of antibiotic resistance. Clin Infect Dis 1997; 24 (Suppl 1): S9–16.
3. Blower SM, Gerberding JL. Understanding, predicting and controlling the emergence of drug-resistant tuberculosis: a theoretical framework. J Mol Med 1998; 76: 624–36.
4. Austin DJ, Kristinsson KG, Anderson RM. The relationship between the volume of antimicrobial consumption in human communities and the frequency of resistance. Proc Natl Acad Sci USA 1999; 96: 1152–6.
5. Bonhoeffer S, Lipsitch M, Levin BR. Evaluating treatment protocols to prevent antibiotic resistance. Proc Natl Acad Sci U S A 1997; 94: 12106–11.
6. Austin DJ, Bonten MJ, Weinstein RA, Slaughter S, Anderson RM. Vancomycin-resistant enterococci in intensive-care hospital settings: transmission dynamics, persistence, and the impact of infection control programs. Proc Natl Acad Sci USA 1999; 96: 6908–13.
7. Austin DJ, Anderson RM. Studies of antibiotic resistance within the patient, hospitals and the community using simple mathematical models. Philos Trans R Soc Lond B Biol Sci 1999; 354: 721–38.
8. Lipsitch M, Bergstrom CT, Levin BR. From the cover: the epidemiology of antibiotic resistance in hospitals: paradoxes and prescriptions. Proc Natl Acad Sci USA 2000; 97: 1938–43.
9. Kristinsson KG. Effect of antimicrobial use and other risk factors on antimicrobial resistance in pneumococci. Microb Drug Resist 1997; 3: 117–23.
10. Seppala H, Klaukka T, Vuopio-Varkila J, Muotiala A, Helenius H, Lager K, Huovinen P. The effect of changes in the consumption of macrolide antibiotics on erythromycin resistance in group A streptococci in Finland. Finnish Study Group for Antimicrobial Resistance [see comments]. N Engl J Med 1997; 337: 441–6.
11. Levin BR. Conditions for the existence of R-plasmids in bacterial populations. In: Mitsuhashi S, Rosival L, Krcmery V (eds.) Fourth International Symposium on Antibiotic Resistance. Berlin: Aviceum Prague; Springer-Verlag, 1980: 197–202.
12. Taddei F, Matic I, Godelle B, Radman M. To be a mutator, or how pathogenic and commensal bacteria can evolve rapidly. Trends Microbiol 1997; 5: 427–8; discussion 428–9.

13. LeClerc JE, Li B, Payne WL, Cebula TA. High mutation frequencies among *Escherichia coli* and *Salmonella* pathogens. Science 1996; 274: 1208–11.
14. Oliver A, Canton R, Campo P, Baquero F, Blazquez J. High frequency of hypermutable *Pseudomonas aeruginosa* in cystic fibrosis lung infection. Science 2000; 288: 1251–4.
15. Mao EF, Lane L, Lee J, Miller JH. Proliferation of mutators in a cell population. J Bacteriol 1997; 179: 417–22.
16. Andersson DI, Slechta ES, Roth JR. Evidence that gene amplification underlies adaptive mutability of the bacterial *lac* operon. Science 1998; 282: 1133–5.
17. Edlund T, Normark S. Recombination between short DNA homologies causes tandem duplication. Nature 1981; 292: 269–271.
18. Andersson DI, Levin BR. The biological cost of antibiotic resistance. Curr Opin Microbiol 1999; 2: 489–93.
19. Nissinen A, Gronroos P, Huovinen P et al. Development of beta-lactamase-mediated resistance to penicillin in middle-ear isolates of *Moraxella catarrhalis* in Finnish children, 1978–1993. Clin Infect Dis 1995; 21: 1193–6.
20. Reynolds M. Compensatory evolution in rifampin-resistant *E. coli*. Genetics (in press).
21. Kurland CG, Hughes D, Ehrenberg M. The limitations of translational accuracy. In: Neidhardt FC (ed.) *Escherichia coli* and *Salmonella*: Cellular and Molecular Biology. Washington, DC: ASM Press, 1996: 979–1004.
22. Johanson U, Aevarsson A, Liljas A, Hughes D. The dynamic structure of EF-G studied by fusidic acid resistance and internal revertants. J Mol Biol 1996; 258: 420–32.
23. Compeau G, Al-Achi BJ, Platsouka E, Levy SB. Survival of rifampin-resistant mutants of *Pseudomonas fluorescens* and *Pseudomonas putida* in soil systems. Appl Environ Microbiol 1988; 54: 2432–8.
24. Jin DJ, Gross CA. Characterization of the pleiotropic phenotypes of rifampin-resistant *rpoB* mutants of *Escherichia coli*. J Bacteriol 1989; 171: 5229–31.
25. Cohan FM, King EC, Zawadzki P. Amelioration of the deleterious pleiotropic effects of an adaptive mutation in *Bacillus subtilis*. Evolution 1994; 48: 81–95.
26. Heym B, Stavropoulos E, Honore N et al. Effects of overexpression of the alkyl hydroperoxide reductase AhpC on the virulence and isoniazid resistance of *Mycobacterium tuberculosis*. Infect Immun 1997; 65: 1395–401.
27. Ordway DJ, Sonnenberg MG, Donahue SA, Belisle JT, Orme IM. Drug-resistant strains of *Mycobacterium tuberculosis* exhibit a range of virulence for mice. Infect Immun 1995; 63: 741–3.
28. Wilson TM, de Lisle GW, Collins DM. Effect of inhA and katG on isoniazid resistance and virulence of *Mycobacterium bovis*. Mol Microbiol 1995; 15: 1009–15.
29. Schrag SJ, Perrot V. Reducing antibiotic resistance [letter]. Nature 1996; 381: 120–1.
30. Fermer C, Swedberg G. Adaptation to sulfonamide resistance in *Neisseria meningitidis* may have required compensatory changes to retain enzyme function: kinetic analysis of dihydropteroate synthases from *N. meningitidis* expressed in a knockout mutant of *Escherichia coli.* J Bacteriol 1997; 179: 831–7.
31. Schrag SJ, Perrot V, Levin BR. Adaptation to the fitness costs of antibiotic resistance in *Escherichia coli*. Proc R Soc Lond B Biol Sci 1997; 264: 1287–91.
32. Björkman J, Hughes D, Andersson DI. Virulence of antibiotic-resistant *Salmonella typhimurium*. Proc Natl Acad Sci USA 1998; 95: 3949–53.
33. Dykes GA, Hastings JW. Fitness costs associated with class IIa bacteriocin resistance in *Listeria monocytogenes* B73. Lett Appl Microbiol 1998; 26: 5–8.
34. Zhongming L, Kelley C, Collins F, Rouse D, Morris S. Expression of *katG* in *Mycobacterium tuberculosis* is associated with growth and persistence in mice and guinea pigs. J Infect Dis 1997; 177: 1030–35.
35. Billington OJ, McHugh, TD, Gillespie SH. Physiological cost of rifampin resistance induced in vitro in *Mycobacterium tuberculosis*. Antimicrob Agents Chemother 1999; 43: 1866–9.
36. Björkman J, Samuelsson P, Andersson DI, Hughes D. Novel ribosomal mutations affecting translational accuracy, antibiotic resistance and virulence of *Salmonella typhimurium*. Mol Microbiol 1999; 31: 53–8.
37. Björkman J, Nagaev I, Berg OG, Hughes D, Andersson DI. Effects of environment on compensatory mutations to ameliorate costs of antibiotic resistance. Science 2000; 287: 1479–82.
38. Godwin D, Slater JH. The influence of the growth environment on the stability of a drug resistance plasmid in *Escherichia coli* K12. J Gen Microbiol 1979; 111: 201–10.
39. Zund P, Lebek G. Generation time-prolonging R plasmids: correlation between increases in the generation time of *Escherichia coli* caused by R plasmids and their molecular size. Plasmid 1980; 3: 65–9.
40. Helling RB, Kinney T, Adams J. The maintenance of plasmid-containing organisms in populations of *Escherichia coli*. J Gen Microbiol 1981; 123: 129–41.
41. Lee SW, Edlin G. Expression of tetracycline resistance in pBR322 derivatives reduces the reproductive fitness of plasmid-containing *Escherichia coli*. Gene 1985; 39: 173–80.
42. Bouma JE, Lenski RE. Evolution of a bacteria/plasmid association. Nature 1988; 335: 351–2.
43. Caldwell BA, Ye C, Griffiths RP, Moyer CL, Morita RY. Plasmid expression and maintenance during long-term starvation-survival of bacteria in well water. Appl Environ Microbiol 1989; 55: 1860–4.
44. Leonhardt H, Alonso JC. Parameters affecting plasmid stability in *Bacillus subtilis*. Gene 1991; 103: 107–11.
45. Modi RI, Adams J. Coevolution in bacteria-plasmid populations. Evolution 1991; 45: 656–67.
46. Modi RI, Wilke CM, Rosenzweig RF, Adams J. Plasmid macro-evolution: selection of deletions during adaptation in a nutrient-limited environment. Genetica 1991; 84: 195–202.
47. McDermott PJ, Gowland P, Gowland PC. Adaptation of *Escherichia coli* growth rates to the presence of pBR322. Lett Appl Microbiol 1993; 17: 139–43.
48. Modi RI, Castilla LH, Puskas-Rozsa S, Helling RB, Adams J. Genetic changes accompanying increased fitness in evolving populations of *Escherichia coli*. Genetics 1992; 130: 241–9.
49. Lenski RE, Simpson SC, Nguyen TT. Genetic analysis of a plasmid-encoded, host genotype-specific enhancement of bacterial fitness. J Bacteriol 1994; 176: 3140–7.
50. Valenzuela MS, Ikpeazu EV, Siddiqui KA. *E. coli* growth inhibition by a high copy number derivative of plasmid pBR322. Biochem Biophys Res Commun 1996; 219: 876–83.
51. Sherman DR, Mdluli K, Hickey MJ, Arain TM, Morris SL, Barry CE 3rd, Stover CK. Compensatory *ahpC* gene expression in isoniazid-resistant *Mycobacterium tuberculosis*. Science 1996; 272: 1641–3.
52. Akaike T, Maeda H. Viral mutation and evolution accelerated by nitric oxide-induced oxidative stress. In: Biology of Nitric Oxide: 6th International Meeting (Stockholm, Sweden) 1999.

53. Böttger EC, Springer B, Pletschette M, Sander P. Fitness of antibiotic-resistant microorganisms and compensatory mutations [letter]. Nat Med 1998; 4: 1343–4.

54. Levin BR, Perrot V, Walker NM. Compensatory mutations, antibiotic resistance and the population genetics of adaptive evolution in bacteria. Genetics 2000; 154: 985–997.

